Leak Test of Gearbox Housing

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Document type	Work Instruction	**Status**	Current
Effective date	2026-06-01	**Owner**	Quality Engineering
Classification	Internal		

1 Purpose

Every assembled gearbox housing is leak tested with the pressure decay method on leak test bench LT-2 before it goes to the end-of-line test.

2 Preconditions

The leak test may only be started after the sealant has cured for at least 30 minutes (see WI-ASM-014). The housing temperature must be between 15 and 35 degrees C. Warm parts give false results.

3 Daily master part check

At the start of every shift, test the master part with the calibrated test leak of 1.5 cm3/min. The bench must show a leak rate between 1.3 and 1.7 cm3/min. If not, the bench is blocked and maintenance has to be called. Record the result in the MES.

4 Test parameters

Parameter	Value
Test medium	Dry compressed air
Test pressure	500 mbar (0.5 bar)
Fill time	5 s
Settle time	10 s
Measuring time	8 s
Reject limit	2.0 cm3/min

5 Procedure

1 Place the housing on the fixture and close the sealing adapters.

2 Start the test with the two-hand switch.

3 Green light: the part passed. Remove it and put it on the conveyor to the end-of-line test.

4 Red light: the part failed. Check the sealing adapters for damage or dirt and test the part once more.

5 If the part fails a second time, attach a red tag, put it in the quarantine area and create a nonconformity report (QM-NCR-001). Do not test it a third time.

6 Records

The bench saves pressure curve and leak rate of every part with the serial number in the MES.

7 Revision history

Rev	Date	Change
C	2025-03-10	Settle time 8 -> 10 s
D	2026-06-01	Cure time before test 20 -> 30 min (ECN-2026-044)